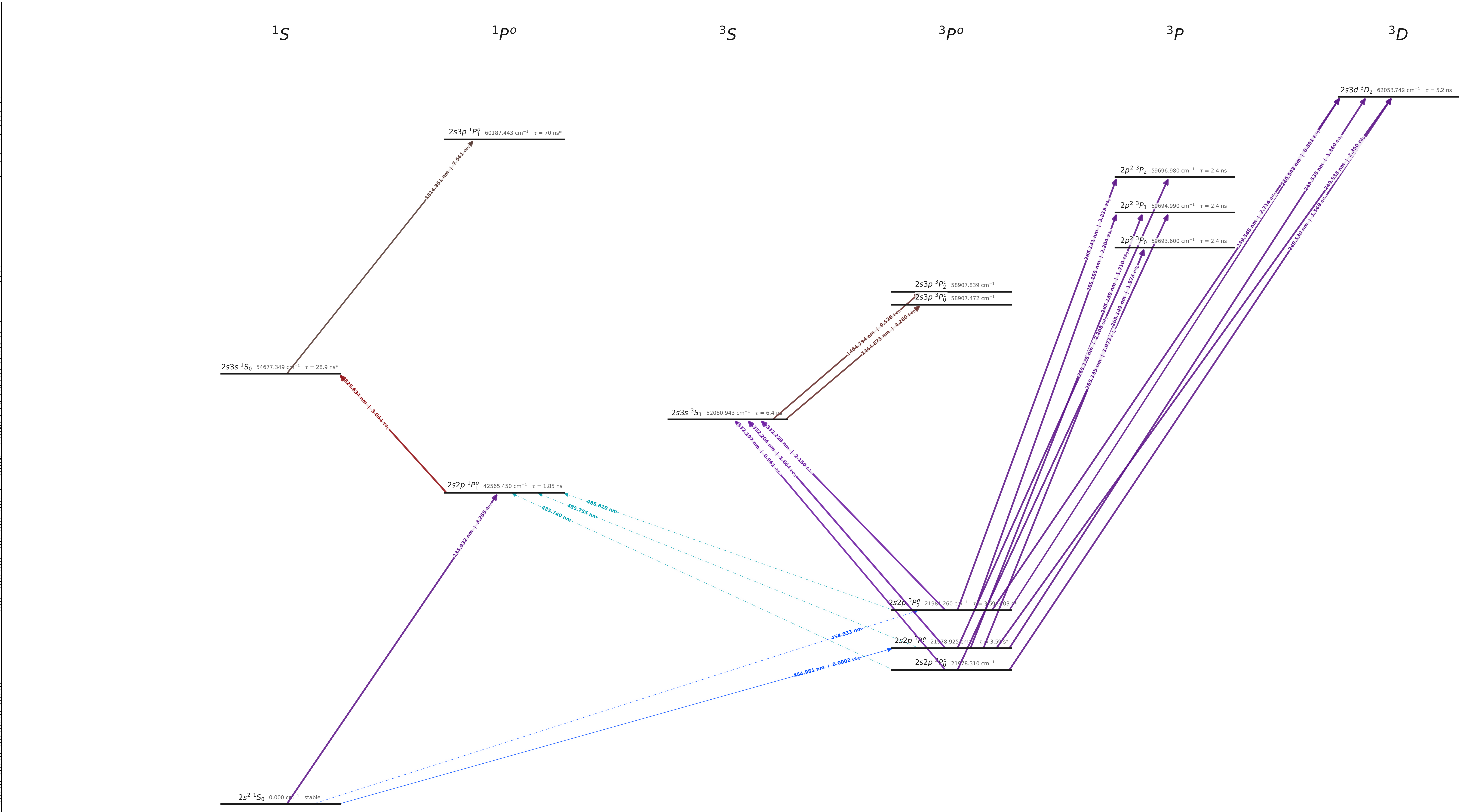


Beryllium-9

⁹Be (Be I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm: 15 levels, 24 lines.

Be⁺ ionisation limit 75192.64 cm⁻¹ = 9.32270 eV (132.992 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er J' (eao)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$2s^2\ ^1S_0 - 2s2p\ ^1P_1^o$	234.9323	234.8604	1276.08008(5)	3.255	5.520e+08	86.03	100	234.9 nm resonance line
$2s^2\ ^1S_0 - 2s2p\ ^3P_0^o$	454.9813	454.8538	658.9116	0.000	4.200e-01	4.43e-08	100	454.9 nm intercombination line
$2s2p\ ^1P_1^o - 2s3s\ ^1S_0$	825.6342	825.4073	363.10565(13)	3.064	3.380e+07	5.517	97.5	825.6 nm

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁹Be hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$2s2p\ ^3P_1^o$	-139.373(12)	-0.75(4)	F=1/2: +347.49 F=3/2: +140.13 F=5/2: -209.25
$2s2p\ ^3P_2^o$	-124.5368(17)	1.429(8)	F=1/2: +561.67 F=3/2: +373.61 F=5/2: +61.38 F=7/2: -373.25
$2s2p\ ^1P_1^o$	-13.836	0.845	F=1/2: +35.65 F=3/2: +12.99 F=5/2: -20.54
$2s3s\ ^3S_1$	-332.250	-	F=1/2: +830.62 F=3/2: +332.25 F=5/2: -498.38
$2p^2\ ^3P_1$	18.101	0.734	F=1/2: -44.34 F=3/2: -18.84 F=5/2: +27.34
$2p^2\ ^3P_2$	3.386	-1.468	F=1/2: -16.52 F=3/2: -10.16 F=5/2: -0.78 F=7/2: +9.79
$2s3p\ ^1P_1^o$	-6.704	0.427	F=1/2: +17.29 F=3/2: +6.28 F=5/2: -9.95
$2s3d\ ^3D_1$	150.600	0.016	F=1/2: -376.48 F=3/2: -150.62 F=5/2: +225.90

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$2s^2\ ^1S_0 - 2s2p\ ^1P_1^o$	234.932	11-9	16015.29(3)

Shift = ν(first isotope) – ν(second isotope).

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J'|$ in eao

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: NIST energy and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

